

# Ritankar Saha

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## EDUCATION

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| <b>Indian Institute of Information Technology, Design and Manufacturing Jabalpur</b> | 3rd Year          |
| • <i>Bachelor of Technology - Computer Science and Engineering</i>                   | <i>2023- 2027</i> |
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## SKILLS SUMMARY

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- **Languages:** Java, C++, C(Sharp), Python, Typescript, JavaScript, GoLang, Rust, Solidity
  - **Data Science:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, CNN, NLP
  - **Frameworks:** Spring Boot, Django, React.js, Next.js, Express.js, Kafka, RabbitMQ, GraphQL, Prisma
  - **Databases:** CouchDB, MariaDB, MongoDB, SQL, PostgreSQL, AWS-DynamoDB
  - **Cloud and DevOps:** Amazon AWS, Kubernetes, Docker, Prometheus, Grafana, Kubestellar, AgroCD
  - **Relevant CourseWork:** DBMS, Operating Systems, Computer Networks, Algorithms, Data Structures
  - **Soft Skills:** Leadership, Public Speaking, Information management, English Proficiency, Product Management
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## EXPERIENCE

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| <b>Google Summer of Code 2025 - Siemens(SW360 and FOSSology)</b>                                                                                                                                                                                                                                                                                                   | Open Source Developer     |
| • <i>Software Developer - Backend and Spring Developer</i>                                                                                                                                                                                                                                                                                                         | <i>May 2025 - Present</i> |
| ◦ Migrated <b>SW360-FOSSology</b> integration to <b>RESTful OpenAPI v2</b> , enabling 1-step multipart uploads with auto-scanning and upload ID return, reducing manual steps by <b>60%</b> and improving API response time by <b>90%</b> .                                                                                                                        |                           |
| ◦ Built <b>Checksum-Driven Reuse</b> via <b>SHA-1/MD5</b> lookup, added user-configurable agent flags ( <b>Nomos, Monk, Bucket</b> ) for selective scans, and automated <b>SPDX/CycloneDX/Tag-Value multi-format</b> report generation, improving overall scan efficiency by <b>80%</b> .                                                                          |                           |
| ◦ <b>Tech Stack:</b> Java, Spring Boot, Thrift, REST APIs, CouchDB, Apache Tomcat, Maven, Docker                                                                                                                                                                                                                                                                   |                           |
| <b>Summer of Bitcoin 2025 - Blockcore(Angor and Angor Hub)</b>                                                                                                                                                                                                                                                                                                     | Open Source Developer     |
| • <i>Blockchain Developer - Elements(Liquid Blockchain) and .NET Developer</i>                                                                                                                                                                                                                                                                                     | <i>May 2025 - Present</i> |
| ◦ Built <b>LiquidService</b> using <b>WASM-based LWK</b> for browser-side confidential wallet with <b>10+ features</b> (addresses, balances, PSETs, CTs), reducing backend load by <b>80%</b> via <b>Nostr Protocol</b> . Integrated with Liquid Mempool using Electrs and ran in parallel with <b>BitcoinService</b> supporting <b>100% client-side signing</b> . |                           |
| ◦ Added <b>Multi-Asset and Confidentiality</b> support (L-BTC, USDT, BTC), <b>Taproot scripts, multi-party PSET signing</b> , and tested <b>BCIP-0005</b> funding protocol on tesnet/mainnet.                                                                                                                                                                      |                           |
| ◦ <b>Tech Stack:</b> C(Sharp), Electrs, Fulcrum, Rust, Blockchain, Bitcoin, Liquid Network, Elements                                                                                                                                                                                                                                                               |                           |
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## PROJECTS

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- **GitHubHub: Golang, Shell, MCP Server Protocol, AWS**  
Developed a terminal-based tool integrating seamlessly with any app to simplify GitHub CLI usage, enabling advanced Git operations without memorizing commands. Built with **MCP protocol**; adopted by **SocialCalc** and **EtherCalc** under **Taiwan's** national research infra, indirectly impacting **300K+** users. Used by **80+ developers**.  
**GitHub:** [github.com/ritankarsaha/GitHubber](https://github.com/ritankarsaha/GitHubber) **Live Demo:** [github.com/ritankarsaha/GitHubber/releases](https://github.com/ritankarsaha/GitHubber/releases)
  - **Decentralised News Application: Golang, Next.js, Python, GenAI, MongoDB, PostgreSQL**  
Built a **Decentralized Web3** News platform with a **Notion-style** workspace, integrating custom-made **NewsToken** for incentivized publishing and 2 custom AI models for content curation, misinformation detection. Secured 10,000 dollars in cloud/Web3 credits to scale.  
**GitHub:** [github.com/ritankarsaha/Decentralised-News-Application](https://github.com/ritankarsaha/Decentralised-News-Application)
  - **Advanced Trading Analytics Platform: Python, Flask, FastAPI, React, WebSockets**  
Developed a **full-stack Trading Platform** for Safe Market Forecasting, Forex Trading, and return prediction. Includes a real-time execution engine, **Visual Strategy Builder** (20+ indicators), and comprehensive backtesting with live monitoring. Powered 1,000+ strategy simulations across global markets.  
**GitHub:** [github.com/ritankarsaha/backtesting-platform](https://github.com/ritankarsaha/backtesting-platform) **Live Demo:** Demonstration of the Platform
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## ACHIEVEMENTS

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- Contributed **60+ PRs** to open-source projects like **SW360, FOSSology, KubeStellar, KubeBurner** and **Podman (Red-Hat Enterprise)**.
  - Earned **Machine Learning Specialization** Certificate from Andrew Ng's Stanford Courses.
  - Ranked **5th/115** at EIBS (East India Blockchain Summit) for building a Decentralized Healthcare DApp and **1st place** in Zero-Trust Hackathon with Terraform-based IaaS solution on AWS.
  - Solved **800+** problems on LeetCode, CodeChef, Codeforces, etc.
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## LEADERSHIP AND VOLUNTEERING

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- Co-coordinator of the Programming Club; conducted **18+ sessions** for **600+ students** on DSA and competitive programming.